

Nyx Haile

AI Systems & Research Engineer

xyn@mit.edu | LinkedIn | GitHub | Open to remote work and relocation

EDUCATION

Massachusetts Institute of Technology

2026

B.S. Computer Science, B.S. Mathematics, Minor in Economics

Cambridge, MA

EXPERIENCE

Undergraduate Researcher

June 2025 – Present

Weiss Laboratory for Synthetic Biology

Cambridge, MA

- Built LLM-assisted generative design workflows for biologically constrained vascular microarchitectures using graph algorithms, spatial optimization, and formal constraints.
- Configured and benchmarked high-performance computing workflows for mesh generation, validation, and experimental export across platforms.

Research Mentor

September 2023 – Present

The SHED

Cambridge, MA

- Mentored interdisciplinary research in biocomputing, HCI, and personal transportation, while training lab staff on equipment and safety procedures.

AI Developer

December 2022 – July 2023

Vana

Remote

- Organized the \$25K Vana x MIT Generative AI Hackathon, built its Python and JavaScript participant infrastructure, and shipped a Swift client for the Vana API.

SELECTED PROJECTS

howtowlol | Python, PyTorch, CUDA/AMP, Redis, Svelte/SvelteKit *World model for post-game teaching*

- Designed and trained a Dreamer-lineage RSSM world model with static context, player priors, event sequences, outcome, next-event, next-frame, and next-decision objectives.
- Trained Plan B end-to-end on a 51k-game remote-GPU corpus with CUDA/AMP; reached outcome AUC@15 of about 0.884 on game-cold and 0.841 on player-cold holdouts, with a clean leak probe at about 0.531.
- Built and evaluated a 1.16M-by-544 anchor retrieval index with GPU kNN across game-cold and player-cold holdouts, comparing random, static, and frame-feature baselines.

Forge | Rust, Linux/systemd, RPC

Local-first agent control plane

- Built a Rust daemon, CLI, hooks, scheduler, and worker/runtime orchestration layer for coordinating agent work across projects.
- Specified typed model-trust, prompt-gate, resource-pressure, and worker-terminal contracts, with systemd sandboxing for bounded local execution.
- Implemented evaluation, audit, provenance, and replay surfaces through structured daemon artifacts, logs, metadata, and tests.

Neuro | Python, NumPy

Exploratory adaptive sparse neural core

- Explored resource-bounded induction with sampled activation, copula/co-firing statistics, sparse pruning, and graph rewiring.

TECHNICAL SKILLS

Languages: Rust, Python, C/C++, SQL **ML:** PyTorch, NumPy, GPU-aware training

Systems/Data: Linux/systemd, Docker, Redis, Postgres, Git **Web:** Svelte/SvelteKit, FastAPI/Flask